

Table 1: Detailed instance-level results of MMO_b , BMA, RS for Next Release Problem, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). Blue cells denote the best Scott-Knott ESD ranks, Yellow cells denote the algorithm (among the three) with the smallest mean in that dataset.

Instance	Rank			Mean $\pm$ Std		
	MMO_b	BMA	RS	MMO_b	BMA	RS
nrp-e1	1.0	2.0	3.0	0.002 ± 0.002	0.468 ± 0.027	0.988 ± 0.010
nrp-e2	1.0	2.0	3.0	0.001 ± 0.002	0.468 ± 0.035	0.964 ± 0.024
nrp-e3	1.0	2.0	3.0	0.000 ± 0.000	0.450 ± 0.029	0.957 ± 0.021
nrp-e4	1.0	2.0	3.0	0.000 ± 0.001	0.442 ± 0.041	0.967 ± 0.031
nrp-g1	1.0	2.0	3.0	0.000 ± 0.000	0.431 ± 0.030	0.973 ± 0.025
nrp-g2	1.0	2.0	3.0	0.000 ± 0.000	0.362 ± 0.048	0.973 ± 0.026
nrp-g3	1.0	2.0	3.0	0.000 ± 0.000	0.389 ± 0.046	0.965 ± 0.030
nrp-g4	1.0	2.0	3.0	0.000 ± 0.000	0.375 ± 0.039	0.967 ± 0.031
nrp-m1	1.0	2.0	3.0	0.001 ± 0.002	0.555 ± 0.040	0.974 ± 0.018
nrp-m2	1.0	2.0	3.0	0.002 ± 0.001	0.531 ± 0.048	0.978 ± 0.018
nrp-m3	1.0	2.0	3.0	0.001 ± 0.001	0.556 ± 0.028	0.978 ± 0.018
nrp-m4	1.0	2.0	3.0	0.001 ± 0.001	0.511 ± 0.029	0.977 ± 0.013
nrp1	1.0	2.0	3.0	0.000 ± 0.000	0.056 ± 0.029	0.938 ± 0.050
nrp2	1.0	2.0	3.0	0.031 ± 0.020	0.528 ± 0.060	0.972 ± 0.014
nrp3	1.0	2.0	3.0	0.003 ± 0.002	0.292 ± 0.029	0.979 ± 0.017
nrp4	1.0	2.0	3.0	0.000 ± 0.000	0.340 ± 0.029	0.988 ± 0.009
nrp5	1.0	2.0	3.0	0.000 ± 0.000	0.265 ± 0.023	0.987 ± 0.010
Average	1.00	2.00	3.00	0.003 ± 0.002	0.413 ± 0.036	0.972 ± 0.021

Table 2: Detailed instance-level results of MMO_b, PromiseTune, and RS for Software Configuration Tuning, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). $\times$ indicates that the algorithm failed to obtain valid solutions on that instance. The color coding and ranking conventions used herein are consistently applied to all subsequent tables.

Instance	Rank			Mean $\pm$ Std		
	MMO _b	PromiseTune	RS	MMO _b	PromiseTune	RS
dnn_adiac	1.0	1.0	2.0	0.328 $\pm$ 0.354	0.522 $\pm$ 0.490	0.820 $\pm$ 0.142
dnn_adiac_reverse	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.100 $\pm$ 0.316	0.094 $\pm$ 0.034
dnn_coffee	1.0	3.0	2.0	0.000 $\pm$ 0.000	0.800 $\pm$ 0.310	0.214 $\pm$ 0.345
dnn_coffee_reverse	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.552 $\pm$ 0.281
dnn_dsr	1.0	3.0	2.0	0.004 $\pm$ 0.011	0.415 $\pm$ 0.398	0.113 $\pm$ 0.232
dnn_dsr_reverse	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.217 $\pm$ 0.286	0.378 $\pm$ 0.271
dnn_sa	1.0	2.0	3.0	0.023 $\pm$ 0.073	0.223 $\pm$ 0.416	0.622 $\pm$ 0.125
dnn_sa_reverse	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.573 $\pm$ 0.268
llvm	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.600 $\pm$ 0.298
llvm_reverse	1.0	2.0	3.0	0.134 $\pm$ 0.072	0.405 $\pm$ 0.243	0.596 $\pm$ 0.316
lrzip	1.0	2.0	3.0	0.167 $\pm$ 0.127	0.424 $\pm$ 0.123	0.649 $\pm$ 0.207
lrzip_reverse	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.001 $\pm$ 0.002	0.595 $\pm$ 0.314
mariadb	1.0	1.0	2.0	0.000 $\pm$ 0.000	0.264 $\pm$ 0.425	0.360 $\pm$ 0.466
mariadb_reverse	1.0	3.0	2.0	0.000 $\pm$ 0.000	0.318 $\pm$ 0.454	0.053 $\pm$ 0.111
mongodb	1.0	2.0	2.0	0.032 $\pm$ 0.101	0.000 $\pm$ 0.000	0.292 $\pm$ 0.350
mongodb_reverse	1.0	2.0	2.0	0.026 $\pm$ 0.081	0.026 $\pm$ 0.081	0.314 $\pm$ 0.404
storm_rs	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.525 $\pm$ 0.458
storm_rs_reverse	1.0	3.0	2.0	0.000 $\pm$ 0.000	0.247 $\pm$ 0.337	0.227 $\pm$ 0.173
storm_wc	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.406 $\pm$ 0.295
storm_wc_reverse	1.0	2.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.738 $\pm$ 0.240
trimesh	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
trimesh_reverse	1.0	3.0	2.0	0.015 $\pm$ 0.012	0.613 $\pm$ 0.226	0.337 $\pm$ 0.113
vp9	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.043 $\pm$ 0.070	0.346 $\pm$ 0.284
vp9_reverse	1.0	1.0	2.0	0.043 $\pm$ 0.090	0.207 $\pm$ 0.298	0.152 $\pm$ 0.209
x264	1.0	2.0	1.0	0.500 $\pm$ 0.000	$\times$	0.500 $\pm$ 0.000
x264_reverse	1.0	2.0	1.0	0.500 $\pm$ 0.000	$\times$	0.500 $\pm$ 0.000
Average	1.00	2.04	2.08	0.087 $\pm$ 0.035	0.282 $\pm$ 0.172	0.425 $\pm$ 0.228

Table 3: Detailed instance-level results of MMO_b, GA, and RS for Software Effort Estimation, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better).

Instance	Rank			Mean $\pm$ Std		
	MMO _b	GA	RS	MMO _b	GA	RS
china_fold1	1.0	2.0	3.0	0.078 $\pm$ 0.053	0.160 $\pm$ 0.135	0.709 $\pm$ 0.168
china_fold2	1.0	2.0	3.0	0.021 $\pm$ 0.025	0.056 $\pm$ 0.054	0.804 $\pm$ 0.189
china_fold3	1.0	2.0	3.0	0.004 $\pm$ 0.005	0.016 $\pm$ 0.028	0.586 $\pm$ 0.248
desharnais_fold1	1.0	2.0	3.0	0.145 $\pm$ 0.136	0.396 $\pm$ 0.244	0.772 $\pm$ 0.154
desharnais_fold2	1.0	2.0	3.0	0.035 $\pm$ 0.027	0.264 $\pm$ 0.318	0.279 $\pm$ 0.074
desharnais_fold3	1.0	2.0	3.0	0.117 $\pm$ 0.088	0.332 $\pm$ 0.090	0.788 $\pm$ 0.149
finnish_fold1	2.0	1.0	2.0	0.064 $\pm$ 0.044	0.082 $\pm$ 0.072	0.541 $\pm$ 0.231
finnish_fold2	1.0	2.0	3.0	0.079 $\pm$ 0.092	0.172 $\pm$ 0.113	0.782 $\pm$ 0.141
finnish_fold3	1.0	2.0	3.0	0.016 $\pm$ 0.012	0.031 $\pm$ 0.026	0.712 $\pm$ 0.159
maxwell_fold1	2.0	1.0	3.0	0.042 $\pm$ 0.025	0.035 $\pm$ 0.028	0.907 $\pm$ 0.055
maxwell_fold2	2.0	1.0	2.0	0.055 $\pm$ 0.030	0.060 $\pm$ 0.045	0.943 $\pm$ 0.034
maxwell_fold3	2.0	1.0	3.0	0.075 $\pm$ 0.042	0.057 $\pm$ 0.034	0.899 $\pm$ 0.070
miyazaki_fold1	1.0	2.0	3.0	0.003 $\pm$ 0.002	0.006 $\pm$ 0.005	0.500 $\pm$ 0.242
miyazaki_fold2	1.0	2.0	3.0	0.002 $\pm$ 0.002	0.006 $\pm$ 0.003	0.482 $\pm$ 0.252
miyazaki_fold3	1.0	2.0	3.0	0.003 $\pm$ 0.001	0.005 $\pm$ 0.003	0.436 $\pm$ 0.260
Average	1.27	1.73	2.87	0.049 $\pm$ 0.039	0.112 $\pm$ 0.080	0.676 $\pm$ 0.162

Table 4: Detailed instance-level results of MMO_b, SBWOA, and RS for Software Defect Prediction, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better).

Instance	Rank			Mean $\pm$ Std		
	MMO _b	SBWOA	RS	MMO _b	SBWOA	RS
ant-1.7_J48	1.0	1.0	2.0	0.101 $\pm$ 0.067	0.555 $\pm$ 0.196	0.604 $\pm$ 0.222
ant-1.7_KNN	1.0	1.0	2.0	0.133 $\pm$ 0.067	0.455 $\pm$ 0.121	0.476 $\pm$ 0.228
ant-1.7_LR	1.0	1.0	2.0	0.433 $\pm$ 0.145	0.626 $\pm$ 0.288	0.648 $\pm$ 0.331
ant-1.7_NB	1.0	1.0	2.0	0.187 $\pm$ 0.101	0.457 $\pm$ 0.279	0.442 $\pm$ 0.246
camel-1.6_J48	1.0	2.0	3.0	0.094 $\pm$ 0.071	0.496 $\pm$ 0.175	0.560 $\pm$ 0.267
camel-1.6_KNN	1.0	3.0	2.0	0.149 $\pm$ 0.075	0.636 $\pm$ 0.272	0.537 $\pm$ 0.249
camel-1.6_LR	1.0	1.0	2.0	0.175 $\pm$ 0.101	0.445 $\pm$ 0.277	0.458 $\pm$ 0.334
camel-1.6_NB	1.0	1.0	2.0	0.391 $\pm$ 0.108	0.574 $\pm$ 0.233	0.509 $\pm$ 0.246
ivy-2.0_J48	1.0	1.0	2.0	0.290 $\pm$ 0.142	0.762 $\pm$ 0.141	0.787 $\pm$ 0.134
ivy-2.0_KNN	1.0	1.0	2.0	0.166 $\pm$ 0.087	0.559 $\pm$ 0.190	0.528 $\pm$ 0.227
ivy-2.0_LR	1.0	1.0	2.0	0.147 $\pm$ 0.065	0.453 $\pm$ 0.246	0.477 $\pm$ 0.259
ivy-2.0_NB	1.0	1.0	2.0	0.138 $\pm$ 0.060	0.501 $\pm$ 0.241	0.442 $\pm$ 0.165
jedit-4.0_J48	1.0	2.0	3.0	0.134 $\pm$ 0.083	0.644 $\pm$ 0.206	0.766 $\pm$ 0.208
jedit-4.0_KNN	1.0	1.0	2.0	0.192 $\pm$ 0.083	0.667 $\pm$ 0.181	0.633 $\pm$ 0.172
jedit-4.0_LR	1.0	1.0	2.0	0.129 $\pm$ 0.091	0.574 $\pm$ 0.212	0.592 $\pm$ 0.194
jedit-4.0_NB	1.0	1.0	2.0	0.136 $\pm$ 0.071	0.604 $\pm$ 0.228	0.550 $\pm$ 0.185
lucene-2.4_J48	1.0	1.0	2.0	0.139 $\pm$ 0.076	0.667 $\pm$ 0.190	0.647 $\pm$ 0.126
lucene-2.4_KNN	1.0	2.0	3.0	0.236 $\pm$ 0.128	0.622 $\pm$ 0.206	0.720 $\pm$ 0.227
lucene-2.4_LR	1.0	2.0	3.0	0.434 $\pm$ 0.089	0.567 $\pm$ 0.273	0.636 $\pm$ 0.267
lucene-2.4_NB	1.0	1.0	2.0	0.352 $\pm$ 0.074	0.508 $\pm$ 0.292	0.476 $\pm$ 0.256
poi-3.0_J48	1.0	3.0	2.0	0.142 $\pm$ 0.094	0.778 $\pm$ 0.185	0.583 $\pm$ 0.282
poi-3.0_KNN	1.0	2.0	3.0	0.224 $\pm$ 0.101	0.572 $\pm$ 0.269	0.623 $\pm$ 0.221
poi-3.0_LR	1.0	1.0	2.0	0.242 $\pm$ 0.103	0.466 $\pm$ 0.230	0.476 $\pm$ 0.150
poi-3.0_NB	1.0	3.0	2.0	0.258 $\pm$ 0.118	0.565 $\pm$ 0.215	0.508 $\pm$ 0.186
synapse-1.2_J48	1.0	2.0	3.0	0.102 $\pm$ 0.083	0.555 $\pm$ 0.204	0.661 $\pm$ 0.202
synapse-1.2_KNN	1.0	2.0	3.0	0.137 $\pm$ 0.099	0.581 $\pm$ 0.242	0.605 $\pm$ 0.202
synapse-1.2_LR	1.0	1.0	2.0	0.216 $\pm$ 0.118	0.584 $\pm$ 0.199	0.597 $\pm$ 0.212
synapse-1.2_NB	1.0	1.0	2.0	0.197 $\pm$ 0.117	0.553 $\pm$ 0.242	0.567 $\pm$ 0.217
velocity-1.6_J48	1.0	1.0	2.0	0.075 $\pm$ 0.048	0.516 $\pm$ 0.262	0.509 $\pm$ 0.127
velocity-1.6_KNN	1.0	2.0	3.0	0.141 $\pm$ 0.089	0.439 $\pm$ 0.239	0.478 $\pm$ 0.221
velocity-1.6_LR	1.0	1.0	2.0	0.109 $\pm$ 0.091	0.357 $\pm$ 0.279	0.358 $\pm$ 0.275
velocity-1.6_NB	1.0	3.0	2.0	0.420 $\pm$ 0.082	0.597 $\pm$ 0.280	0.437 $\pm$ 0.200
xalan-2.6_J48	1.0	3.0	2.0	0.101 $\pm$ 0.078	0.492 $\pm$ 0.207	0.420 $\pm$ 0.159
xalan-2.6_KNN	1.0	3.0	2.0	0.134 $\pm$ 0.120	0.631 $\pm$ 0.111	0.521 $\pm$ 0.276
xalan-2.6_LR	1.0	1.0	1.0	0.361 $\pm$ 0.130	0.426 $\pm$ 0.308	0.431 $\pm$ 0.297
xalan-2.6_NB	2.0	1.0	2.0	0.292 $\pm$ 0.121	0.370 $\pm$ 0.278	0.483 $\pm$ 0.269
xerces-1.4_J48	1.0	2.0	1.0	0.226 $\pm$ 0.100	0.534 $\pm$ 0.347	0.316 $\pm$ 0.276
xerces-1.4_KNN	1.0	1.0	2.0	0.181 $\pm$ 0.170	0.604 $\pm$ 0.232	0.568 $\pm$ 0.246
xerces-1.4_LR	1.0	1.0	2.0	0.456 $\pm$ 0.138	0.595 $\pm$ 0.309	0.563 $\pm$ 0.319
xerces-1.4_NB	1.0	3.0	2.0	0.272 $\pm$ 0.152	0.578 $\pm$ 0.338	0.464 $\pm$ 0.338
Average	1.02	1.57	2.15	0.211 $\pm$ 0.098	0.554 $\pm$ 0.236	0.541 $\pm$ 0.231

Table 5: Detailed instance-level results of MMO_b , LS-Sampling-Plus, RS for Software Product Lines Testing, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). Blue cells denote the best Scott-Knott ESD ranks, Yellow cells denote the algorithm (among the three) with the smallest mean in that dataset.

Instance	Rank			Mean $\pm$ Std		
	MMO_b	LS-Sampling-Plus	RS	MMO_b	LS-Sampling-Plus	RS
7z	1.0	2.0	3.0	0.007 $\pm$ 0.006	0.463 $\pm$ 0.176	0.640 $\pm$ 0.218
amazon	1.0	2.0	3.0	0.030 $\pm$ 0.015	0.229 $\pm$ 0.131	0.720 $\pm$ 0.130
berkeleydbc	1.0	3.0	2.0	0.038 $\pm$ 0.030	0.871 $\pm$ 0.154	0.629 $\pm$ 0.143
cocheecologico	1.0	2.0	3.0	0.018 $\pm$ 0.016	0.485 $\pm$ 0.205	0.825 $\pm$ 0.127
counterstrikesimplefeatu	1.0	2.0	3.0	0.032 $\pm$ 0.031	0.686 $\pm$ 0.136	0.846 $\pm$ 0.115
drupal	1.0	2.0	3.0	0.052 $\pm$ 0.022	0.374 $\pm$ 0.071	0.877 $\pm$ 0.073
dssample	1.0	3.0	2.0	0.017 $\pm$ 0.014	0.560 $\pm$ 0.357	0.400 $\pm$ 0.057
dune	1.0	3.0	2.0	0.058 $\pm$ 0.042	0.915 $\pm$ 0.057	0.723 $\pm$ 0.051
electronicdrum	1.0	2.0	3.0	0.107 $\pm$ 0.042	0.778 $\pm$ 0.066	0.842 $\pm$ 0.103
hipacc	1.0	2.0	3.0	0.058 $\pm$ 0.030	0.451 $\pm$ 0.072	0.820 $\pm$ 0.123
javagc	1.0	2.0	3.0	0.034 $\pm$ 0.020	0.447 $\pm$ 0.061	0.890 $\pm$ 0.129
jhipster	1.0	2.0	3.0	0.052 $\pm$ 0.030	0.520 $\pm$ 0.186	0.906 $\pm$ 0.093
lrzip	1.0	2.0	1.0	0.000 $\pm$ 0.000	0.544 $\pm$ 0.385	0.006 $\pm$ 0.020
modeltransformation	1.0	2.0	3.0	0.080 $\pm$ 0.050	0.336 $\pm$ 0.073	0.878 $\pm$ 0.145
polly	1.0	2.0	3.0	0.072 $\pm$ 0.044	0.748 $\pm$ 0.134	0.792 $\pm$ 0.152
smarthomev2.2	1.0	2.0	3.0	0.039 $\pm$ 0.026	0.356 $\pm$ 0.042	0.865 $\pm$ 0.102
splssimuelesnp	1.0	3.0	2.0	0.070 $\pm$ 0.035	0.833 $\pm$ 0.081	0.790 $\pm$ 0.114
videoplayer	1.0	2.0	3.0	0.065 $\pm$ 0.037	0.467 $\pm$ 0.069	0.922 $\pm$ 0.069
vp9	1.0	2.0	3.0	0.087 $\pm$ 0.036	0.562 $\pm$ 0.077	0.768 $\pm$ 0.096
webportal	1.0	2.0	3.0	0.050 $\pm$ 0.033	0.370 $\pm$ 0.034	0.902 $\pm$ 0.061
x264	1.0	3.0	2.0	0.018 $\pm$ 0.028	0.900 $\pm$ 0.062	0.629 $\pm$ 0.062
Average	1.00	2.24	2.67	0.047 $\pm$ 0.028	0.566 $\pm$ 0.125	0.746 $\pm$ 0.104

Table 6: Detailed instance-level results of MMO_b , GP-ILS, RS for Algorithm Scott Knott Final Ranking Software Project Scheduling, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). Blue cells denote the best Scott-Knott ESD ranks, Yellow cells denote the algorithm (among the three) with the smallest mean in that dataset.

Instance	Rank			Mean $\pm$ Std		
	MMO_b	GP-ILS	RS	MMO_b	GP-ILS	RS
10-10-skill-4-5	1.0	2.0	3.0	0.003 $\pm$ 0.002	0.093 $\pm$ 0.022	0.966 $\pm$ 0.027
10-10-skill-6-7	1.0	2.0	3.0	0.005 $\pm$ 0.003	0.061 $\pm$ 0.011	0.972 $\pm$ 0.024
10-15-skill-4-5	1.0	2.0	3.0	0.001 $\pm$ 0.001	0.102 $\pm$ 0.022	0.978 $\pm$ 0.018
10-15-skill-6-7	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.081 $\pm$ 0.013	0.992 $\pm$ 0.005
10-5-skill-4-5	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.012 $\pm$ 0.007	0.975 $\pm$ 0.020
10-5-skill-6-7	1.0	2.0	3.0	0.001 $\pm$ 0.001	0.062 $\pm$ 0.029	0.965 $\pm$ 0.028
20-10-skill-4-5	1.0	2.0	3.0	0.002 $\pm$ 0.001	0.089 $\pm$ 0.033	0.981 $\pm$ 0.014
20-10-skill-6-7	1.0	2.0	3.0	0.001 $\pm$ 0.001	0.069 $\pm$ 0.019	0.987 $\pm$ 0.007
20-15-skill-4-5	1.0	2.0	3.0	0.003 $\pm$ 0.002	0.100 $\pm$ 0.013	0.984 $\pm$ 0.009
20-15-skill-6-7	1.0	2.0	3.0	0.006 $\pm$ 0.006	0.150 $\pm$ 0.032	0.976 $\pm$ 0.023
20-5-skill-4-5	1.0	2.0	3.0	0.003 $\pm$ 0.002	0.142 $\pm$ 0.051	0.961 $\pm$ 0.025
20-5-skill-6-7	1.0	2.0	3.0	0.002 $\pm$ 0.002	0.082 $\pm$ 0.024	0.956 $\pm$ 0.033
30-10-skill-4-5	1.0	2.0	3.0	0.002 $\pm$ 0.001	0.074 $\pm$ 0.033	0.974 $\pm$ 0.014
30-10-skill-6-7	1.0	2.0	3.0	0.003 $\pm$ 0.002	0.133 $\pm$ 0.060	0.984 $\pm$ 0.010
30-15-skill-4-5	1.0	2.0	3.0	0.003 $\pm$ 0.001	0.103 $\pm$ 0.010	0.975 $\pm$ 0.025
30-15-skill-6-7	1.0	2.0	3.0	0.005 $\pm$ 0.004	0.123 $\pm$ 0.025	0.976 $\pm$ 0.020
30-5-skill-4-5	1.0	2.0	3.0	0.004 $\pm$ 0.002	0.252 $\pm$ 0.087	0.953 $\pm$ 0.042
30-5-skill-6-7	1.0	2.0	3.0	0.002 $\pm$ 0.001	0.276 $\pm$ 0.088	0.964 $\pm$ 0.023
Average	1.00	2.00	3.00	0.003 $\pm$ 0.002	0.111 $\pm$ 0.032	0.973 $\pm$ 0.020

Table 7: Detailed instance-level results of MMO_b , GA, and RS for Third Party Library Migration, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better).

Instance	Rank			Mean $\pm$ Std		
	MMO_b	GA	RS	MMO_b	GA	RS
migrationRule10_ALL	2.0	2.0	1.0	0.000 $\pm$ 0.000	0.253 $\pm$ 0.356	0.000 $\pm$ 0.000
migrationRule10_CO	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule10_DS	1.0	2.0	1.0	0.000 $\pm$ 0.000	0.600 $\pm$ 0.516	0.000 $\pm$ 0.000
migrationRule10_MS	2.0	2.0	1.0	0.000 $\pm$ 0.000	0.200 $\pm$ 0.422	0.000 $\pm$ 0.000
migrationRule18_ALL	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.180 $\pm$ 0.306	0.191 $\pm$ 0.140
migrationRule18_CO	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule18_DS	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule18_MS	1.0	3.0	2.0	0.000 $\pm$ 0.000	0.117 $\pm$ 0.315	0.033 $\pm$ 0.070
migrationRule1_ALL	1.0	2.0	3.0	0.021 $\pm$ 0.021	0.234 $\pm$ 0.158	0.833 $\pm$ 0.114
migrationRule1_CO	1.0	2.0	3.0	0.011 $\pm$ 0.016	0.274 $\pm$ 0.224	0.926 $\pm$ 0.076
migrationRule1_DS	2.0	2.0	1.0	0.000 $\pm$ 0.000	0.200 $\pm$ 0.422	0.000 $\pm$ 0.000
migrationRule1_MS	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.029 $\pm$ 0.090	0.871 $\pm$ 0.045
migrationRule2_ALL	1.0	2.0	3.0	0.048 $\pm$ 0.049	0.284 $\pm$ 0.119	0.934 $\pm$ 0.052
migrationRule2_CO	1.0	2.0	3.0	0.079 $\pm$ 0.078	0.260 $\pm$ 0.088	0.952 $\pm$ 0.031
migrationRule2_DS	1.0	2.0	3.0	0.064 $\pm$ 0.041	0.242 $\pm$ 0.113	0.941 $\pm$ 0.040
migrationRule2_MS	1.0	2.0	3.0	0.110 $\pm$ 0.075	0.271 $\pm$ 0.127	0.933 $\pm$ 0.046
migrationRule3_ALL	1.0	1.0	2.0	0.158 $\pm$ 0.243	0.563 $\pm$ 0.206	0.592 $\pm$ 0.246
migrationRule3_CO	1.0	3.0	2.0	0.271 $\pm$ 0.238	0.578 $\pm$ 0.303	0.440 $\pm$ 0.126
migrationRule3_MS	1.0	2.0	3.0	0.050 $\pm$ 0.158	0.250 $\pm$ 0.354	0.500 $\pm$ 0.000
migrationRule4_ALL	1.0	2.0	3.0	0.009 $\pm$ 0.028	0.166 $\pm$ 0.136	0.863 $\pm$ 0.117
migrationRule4_CO	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.100 $\pm$ 0.211	0.800 $\pm$ 0.258
migrationRule4_DS	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.055 $\pm$ 0.172	0.815 $\pm$ 0.188
migrationRule4_MS	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.833 $\pm$ 0.112
migrationRule5_ALL	1.0	1.0	2.0	0.095 $\pm$ 0.210	0.514 $\pm$ 0.290	0.571 $\pm$ 0.150
migrationRule5_CO	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.333 $\pm$ 0.385	0.600 $\pm$ 0.263
migrationRule5_DS	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.551 $\pm$ 0.475	0.903 $\pm$ 0.000
migrationRule5_MS	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.100 $\pm$ 0.316	0.200 $\pm$ 0.422
migrationRule7_ALL	1.0	2.0	3.0	0.133 $\pm$ 0.074	0.306 $\pm$ 0.136	0.897 $\pm$ 0.074
migrationRule7_CO	1.0	2.0	3.0	0.085 $\pm$ 0.062	0.245 $\pm$ 0.157	0.879 $\pm$ 0.081
migrationRule7_DS	1.0	2.0	3.0	0.113 $\pm$ 0.049	0.231 $\pm$ 0.098	0.884 $\pm$ 0.070
migrationRule7_MS	1.0	2.0	3.0	0.125 $\pm$ 0.083	0.481 $\pm$ 0.241	0.894 $\pm$ 0.084
migrationRule8_ALL	1.0	2.0	3.0	0.000 $\pm$ 0.000	0.100 $\pm$ 0.211	0.729 $\pm$ 0.312
migrationRule8_CO	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.044 $\pm$ 0.141	0.478 $\pm$ 0.240
migrationRule8_DS	1.0	1.0	1.0	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000	0.500 $\pm$ 0.000
migrationRule8_MS	2.0	1.0	2.0	0.000 $\pm$ 0.000	0.020 $\pm$ 0.063	0.560 $\pm$ 0.207
Average	1.26	1.71	2.26	0.096 $\pm$ 0.041	0.279 $\pm$ 0.204	0.601 $\pm$ 0.102

Table 8: Detailed instance-level results of MMO_b , HGA, and RS for Workflow Scheduling, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better).

Instance	Rank			Mean $\pm$ Std		
	MMO_b	HGA	RS	MMO_b	HGA	RS
CyberShake_100	1.0	2.0	3.0	0.097 ± 0.038	0.377 ± 0.280	0.569 ± 0.107
CyberShake_1000	2.0	1.0	2.0	0.279 ± 0.089	0.260 ± 0.161	0.925 ± 0.056
CyberShake_30	1.0	3.0	2.0	0.053 ± 0.066	0.317 ± 0.299	0.153 ± 0.190
CyberShake_50	1.0	3.0	2.0	0.039 ± 0.035	0.342 ± 0.257	0.230 ± 0.046
Epigenomics_100	1.0	2.0	3.0	0.170 ± 0.126	0.527 ± 0.275	0.765 ± 0.102
Epigenomics_24	1.0	3.0	2.0	0.163 ± 0.090	0.606 ± 0.208	0.405 ± 0.084
Epigenomics_46	1.0	3.0	2.0	0.045 ± 0.064	0.630 ± 0.271	0.604 ± 0.128
Epigenomics_997	2.0	1.0	3.0	0.349 ± 0.159	0.233 ± 0.188	0.807 ± 0.116
Inspir_100	1.0	2.0	3.0	0.094 ± 0.065	0.482 ± 0.165	0.839 ± 0.120
Inspir_1000	2.0	1.0	2.0	0.215 ± 0.099	0.245 ± 0.189	0.866 ± 0.089
Inspir_30	1.0	2.0	3.0	0.000 ± 0.000	0.165 ± 0.313	0.282 ± 0.268
Inspir_50	1.0	2.0	3.0	0.067 ± 0.066	0.646 ± 0.172	0.844 ± 0.118
Montage_100	1.0	2.0	3.0	0.080 ± 0.066	0.254 ± 0.088	0.725 ± 0.199
Montage_1000	1.0	2.0	3.0	0.130 ± 0.112	0.178 ± 0.154	0.883 ± 0.092
Montage_25	1.0	3.0	2.0	0.022 ± 0.010	0.649 ± 0.153	0.557 ± 0.024
Montage_50	1.0	2.0	3.0	0.051 ± 0.055	0.426 ± 0.235	0.519 ± 0.104
Sipht_100	1.0	3.0	2.0	0.010 ± 0.012	0.306 ± 0.359	0.117 ± 0.065
Sipht_1000	1.0	2.0	3.0	0.067 ± 0.083	0.204 ± 0.156	0.583 ± 0.310
Sipht_30	1.0	3.0	2.0	0.041 ± 0.054	0.754 ± 0.250	0.609 ± 0.136
Sipht_60	1.0	3.0	2.0	0.016 ± 0.014	0.608 ± 0.456	0.092 ± 0.018
Average	1.15	2.25	2.50	0.099 ± 0.065	0.410 ± 0.232	0.569 ± 0.119

Table 9: Detailed instance-level results of MMO_b , HWOA, and RS for Workflow Scheduling(including reverse instances), showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better).

Instance	Rank			Mean $\pm$ Std		
	MMO_b	HWOA	RS	MMO_b	HWOA	RS
workflow_1	1.0	2.0	3.0	0.144 ± 0.103	0.458 ± 0.282	0.788 ± 0.115
workflow_10	1.0	2.0	3.0	0.359 ± 0.158	0.432 ± 0.281	0.811 ± 0.199
workflow_10_reverse	3.0	1.0	2.0	0.785 ± 0.178	0.570 ± 0.235	0.739 ± 0.171
workflow_1_reverse	2.0	1.0	2.0	0.586 ± 0.244	0.648 ± 0.281	0.736 ± 0.132
workflow_2	1.0	2.0	3.0	0.236 ± 0.143	0.293 ± 0.222	0.842 ± 0.161
workflow_2_reverse	2.0	1.0	2.0	0.628 ± 0.235	0.610 ± 0.293	0.843 ± 0.077
workflow_3	1.0	2.0	3.0	0.284 ± 0.206	0.390 ± 0.105	0.814 ± 0.111
workflow_3_reverse	1.0	2.0	3.0	0.660 ± 0.171	0.671 ± 0.315	0.828 ± 0.089
workflow_4	2.0	1.0	2.0	0.238 ± 0.096	0.271 ± 0.234	0.662 ± 0.201
workflow_4_reverse	1.0	3.0	2.0	0.429 ± 0.244	0.688 ± 0.252	0.676 ± 0.175
workflow_5	2.0	1.0	3.0	0.289 ± 0.138	0.240 ± 0.186	0.679 ± 0.255
workflow_5_reverse	2.0	1.0	3.0	0.627 ± 0.207	0.384 ± 0.294	0.724 ± 0.188
workflow_6	1.0	2.0	3.0	0.163 ± 0.130	0.330 ± 0.204	0.731 ± 0.153
workflow_6_reverse	1.0	2.0	1.0	0.453 ± 0.216	0.558 ± 0.202	0.627 ± 0.108
workflow_7	1.0	2.0	3.0	0.264 ± 0.190	0.477 ± 0.201	0.847 ± 0.108
workflow_7_reverse	1.0	2.0	3.0	0.349 ± 0.206	0.481 ± 0.236	0.566 ± 0.081
workflow_8	2.0	1.0	3.0	0.322 ± 0.187	0.249 ± 0.146	0.762 ± 0.146
workflow_8_reverse	2.0	1.0	2.0	0.517 ± 0.325	0.550 ± 0.253	0.736 ± 0.169
workflow_9	2.0	1.0	2.0	0.234 ± 0.181	0.237 ± 0.195	0.735 ± 0.207
workflow_9_reverse	2.0	1.0	2.0	0.533 ± 0.306	0.505 ± 0.284	0.821 ± 0.158
Average	1.55	1.55	2.50	0.405 ± 0.193	0.452 ± 0.235	0.748 ± 0.150